

Solving a Two Country Model using KKT Conditions

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I The Problem

Consider the two-country model: each country $i = 1, 2$ is endowed with L_i units of labor which can be used to produce two goods x and y in non-negative quantities x_i and y_i , respectively. The production of each unit of good x and each unit of good y in country i require fixed amounts a_i and b_i of labor, respectively, so the production possibility constraints are

$$a_i x_i + b_i y_i \leq L_i.$$

The world prices of the goods, p_x and p_y , are independent of the levels of their production in the countries. Assume that the Ricardian conditions hold:

$$\frac{a_1}{b_1} > \frac{p_x}{p_y} > \frac{a_2}{b_2}.$$

- i) Set up the optimization problems
- ii) State the Kuhn Tucker necessary conditions for each problem. Are they sufficient for optimum?
- iii) Suggest an interpretation of Lagrange multipliers
- iv) Prove that, in optimum, each country will completely specialize according to its comparative advantage.

[†]Definitions and example problems from ECON E-520: Optimization Theory in Economic Analysis instructed by Dr. Michael Kaganovich

II Solution

i) Set up the optimization problems

Solution. The optimization problems are, for Country 1,

$$\begin{aligned} \max_{x_1, y_1} & p_x x_1 + p_y y_1 \\ \text{s.t.} & \\ a_1 x_1 + b_1 y_1 & \leq L_1 \end{aligned}$$

For Country 2,

$$\begin{aligned} \max_{x_2, y_2} & p_x x_2 + p_y y_2 \\ \text{s.t.} & \\ a_2 x_2 + b_2 y_2 & \leq L_2 \end{aligned}$$

For $i = 1, 2$

$$\begin{aligned} \max_{x_i, y_i} & p_x x_i + p_y y_i \\ \text{s.t.} & \\ a_i x_i + b_i y_i & \leq L_i \end{aligned}$$

ii) State the Kuhn Tucker necessary conditions for each problem. Are they sufficient for optimum?

Solution. Let $\mathcal{L}$ be the Lagrangian function, and λ_i for $i = 1, 2$ be the Lagrange Multiplier. Then, the Lagrangian for $i = 1, 2$ is

$$\mathcal{L} = p_x x_i + p_y y_i - \lambda_i (a_i x_i + b_i y_i - L_i)$$

So the KKT conditions are

$$\mathcal{L}_{x_i} = p_x - \lambda_i a_i \leq 0, \quad x_i \geq 0 \text{ with CS}$$

$$\mathcal{L}_{y_i} = p_y - \lambda_i b_i \leq 0, \quad y_i \geq 0 \text{ with CS}$$

$$a_i x_i + b_i y_i \leq L_i, \quad \lambda_i \geq 0 \text{ with CS}$$

Recall that one way to show that the KKT conditions for a maximization problem are sufficient requires that the objective function be concave and differentiable, the constraints to be convex and differentiable, and the KKT necessary conditions to hold. So, since all of the constraints are linear, the CQ holds automatically¹. In addition, the linear constraints imply that they are also convex and differentiable. Finally, since the objective function is linear, it is concave and differentiable. Therefore, the KKT conditions are both necessary and sufficient, and there exists a global maximum.

¹Recall that another way to show that the KKT are necessary is that the CQ matrix has full rank

iii) **Suggest an interpretation of Lagrange multipliers**

Solution. The Lagrange multipliers are the shadow price of labor, i.e. how much extra amount of maximized GNP we obtain from 1 extra unit of labor.

iv) **Prove that, in optimum, each country will completely specialize according to its comparative advantage**

Proof. Complete specialization implies that each country will produce only 1 good. So, we must first rule out the cases where both goods are not produced and when both goods are produced by each country. First, consider if $x_i, y_i > 0$. Then, by CS, the F.O.Cs imply that

$$\begin{aligned} p_x = \lambda_i a_i &\implies \frac{p_x}{a_i} = \lambda_i \\ p_y = b_i \lambda_i &\implies \frac{p_y}{b_i} = \lambda_i \end{aligned}$$

so

$$\frac{p_x}{a_i} = \frac{p_y}{b_i}$$

Which contradicts the Ricardian Condition. If $x_i = y_i = 0$, then by CS

$$p_y < \lambda_i b_i \implies \frac{p_y}{b_i} < \lambda_i \implies \lambda_i > 0$$

Since $b_i > 0$ by the Ricardian Conditions. Since, $\lambda_i > 0$ then by CS

$$a_i x_i + b_i y_i = L_i$$

but since we assumed $x_i = 0 = y_i$ then we have that $0 = L_i$ contradiction. We cannot have zero units of labor. So, it must be that only one country can produce one good only. To see what country 1 specializes in, from Ricardian condition one can see that

$$\frac{a_1}{b_1} > \frac{p_x}{p_y} \implies a_1 p_y > p_x b_1 \implies \frac{p_y}{b_1} > \frac{p_x}{a_1}$$

which clearly means that country 1 specializes in good y . To prove this, assume the opposite. If $x_1 > 0$, then by CS we have that (looking into country one's problem)

$$p_x = \lambda_1 a_1 \implies \lambda_1 = \frac{p_x}{a_1}$$

Now, above we proved that if one good is produced, the other must *not* be produced. So, $y_1 = 0$, which by CS gives

$$p_y < \lambda_1 b_1 \implies \frac{p_y}{b_1} < \lambda_1 \implies \frac{p_y}{b_1} < \frac{p_x}{a_1}$$

which is a contradiction with the Ricardian Condition. Therefore, $x_1 = 0$ and since $\lambda_1 > 0$ then by CS

$$a_1 x_1 + b_1 y_1 = L_1 \implies b_1 y_1 = L_1 \implies y_1 = \frac{L_1}{b_1}$$

For country 2, a very similar argument can be made. ■